Sai Srikar Reddy Kolli

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EDUCATION

Masters in Computer Science **Expected May 2027**
University of Missouri, Columbia, MO 4.0 GPA
Bachelors in Computer Science **May 2024**
Indian Institute of Information Technology (IIIT), Sricity 7.68 GPA

TECHNICAL SKILLS

Programming & Databases: Python, R, Java, C#, C++, JavaScript (TypeScript, Node.js), SQL, PostgreSQL, MongoDB, GraphQL
Web & Application Development: ReactJS, Angular, HTML, CSS, Flask, Django, Spring Boot, PyTest, Selenium
Cloud & DevOps: AWS (EC2, S3, IAM, ACM, Lambda, API Gateway, Step Functions, Amplify), Azure DevOps, Docker, Kubernetes, Ansible, Jenkins, GitLab CI, Prometheus, Grafana

WORK EXPERIENCE

University of Missouri Columbia, MO: Graduate Student Assistant, 08/2025 - Present

- Operationalized **IBM Qiskit** Variational Quantum Circuits within a **PyTorch** workflow, utilizing Hilbert space projection to solve complex class separability issues in NISQ-era environments.
- Developing an intelligent document processing pipeline that allows users to upload and query files (PDF, TXT, CSV) using **RAG (Retrieval-Augmented Generation**) techniques.

MAQ Software: Software Engineering 05/2024 - 06/2025

- Analyzed legacy data workflows and refactored **SQL/PySpark** pipelines in **Azure ADF/Databricks**, improving data quality by 10% and enabling cloud-ready architecture for scalable processing.
- Developed end to end workflow diagrams, semantic models, and automated documentation for **ETL** and **Power BI**, reducing onboarding time by 30% and improving reporting consistency across teams.
- Architected scalable ETL pipelines in **Azure Data Factory (ADF**), establishing Linked Services to ingest data from 7 distinct heterogeneous sources into **Azure Data Lake Storage (ADLS**), serving as the central "**OneLake**" repository for downstream analytics.

SP Software (P) Limited: Associate Software Engineering Intern 02/2024 - 04/2024

- Enhanced image processing pipeline by integrating **ESRGAN** algorithms, achieving a 5% improvement in text clarity and legibility, while decreasing image processing time by 15 milliseconds per image.
- Conducted research to explore cutting-edge methods for enhancing text visibility and quality in images using ESRGAN techniques.

PROJECTS

Ticketmaster , *Personal Project* Fall 2025

- Engineered a robust seat reservation engine handling 10,000+ concurrent requests by implementing a hybrid locking strategy. Implemented atomic batch processing for the real-time seat map, eliminating booking conflicts
- Reduced database read load by 60% and **API** latency by ~200ms by implementing a **Write-Through Caching** strategy with **Redis** for real-time seat availability maps.

GraphNet-Sentry , *Research Project* Fall 2025

- Developed a unified deep learning framework capable of processing heterogeneous data modalities, successfully applying **ResNet-50 + Graph Neural Networks (GNN**) to both Network Intrusion Detection and Satellite Image Analysis.
- Engineered a "**Tabular-to-Image**" transformation engine, enabling the model to achieve 99.9% accuracy on the **CICIOT-2023** security dataset by treating network flows as visual patterns.

CERTIFICATIONS

Microsoft Certified: Fabric Analytics Engineer Associate Link
DeepLearning.AI : Neural Networks and Deep Learning Link